Nguyen Quang Dung

AI Engineer

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Profile Summary

My name is Quang Dung Nguyen, a third-year student at the University of Information Technology, Vietnam National University Ho Chi Minh City, majoring in Computer Engineering. I have hands-on experience in developing AI projects for embedded systems, particularly in computer vision, and have contributed to research on prompt design for optimizing large language models. I consider myself a dedicated, hardworking, and reliable individual. I excel both as a team player and as a confident, dependable leader capable of guiding a group effectively. With the experience I have gained and my passion for this field, I believe I can benefit the enterprise.

Education

University of Information Technology - VNU-HCM — Ho Chi Minh City, Viet Nam

Bachelor in Computer Engineering (3th year Undergraduate) — *Oct 2022 — Present*

Cumulative GPA: 3.27/4.0

English level: B2 (6.0 overal Ielts mock test)

High School for the Gifted - VNU-HCM — Ho Chi Minh City, Viet Nam

Specialized in Physics — *Aug 2019 — Jun 2022*

Average GPA: 3.43/4.0

Achievement

digital.auto GenAI Award — *2024 — Present*

- Phase 1's winner.
- Collaborated with another winning team from Stuttgart to integrate two architectural solutions – prompt engineering and RAG AWS knowledge base, develop for Sofware Define Vehicle - SDV - on digital.auto.
- Conducted research and authored scientific reports in collaboration with the Bosch team and members of the Ferdinand-Steinbeis Institute.
- First author of the research paper "Few-Shot Prompting Techniques for Optimizing LLMs in SDV Code Generation".

Experience

AI Engineer Intern

Bosch Global Software Technology - AirBag Department — *Jun. 2025 – Jan. 2026*

- Creating tools and applications manipulating AI to save effort.
- Receive idea and problem from associate, then develop until deployable.
- Full-stack developer, develop both the backend and frontend.
- Have knowledge in both traditional AI model and Agentic AI.

Technical Skills

Languages: C/C++, Python.
Familiar with source version control software: Git.
Knowledgeable in AI topics such as Agent/Agentic AI, object detection, segmentation, and NLP.
Experienced in feature extraction techniques and applying AI on various kind of data.
Skilled in pruning and optimizing deep learning models for efficiency and speed for embedded systems capability.
Familiar with libraries and frameworks such as OpenCV, PyTorch, TensorFlow, scikit-learn.
Capable of prompting analyzing, evaluating, and fine-tuning machine learning models.

Knowing the process of developing a proper tool from idea to finalization stage.
Proficient in writing, speaking, and reading English technical documentation.

Projects

Multiple-choice AutoGrade System 2025

Developed an Android app and embedded server for automated grading of multiple-choice answer sheets:

- Android app captures answer sheet images and sends them to a Raspberry Pi 5 server via HTTP Flask API.
- The server processes images, performs grading, and generates results automatically.
- Grading results are stored and synchronized in Firebase Realtime Database.
- Android app retrieves grading data from Firebase and displays the results in real-time on the user interface.
- Manipulate MQTT, deploy on esp32cam and esp8266 to trigger camera and paper scroller on old version systems.

DistilBERT Model Pruning for Text Classification 2025

Optimized DistilBERT for real-time NLP with pruning techniques:

- Applied head, FFN, and layer pruning to reduce model size and inference time.
- Tested on email spam and news topic classification tasks.
- Reduced size from 255MB to 7.85MB with minimal accuracy loss (<1%).
- Cut inference time from 30s to under 1s while maintaining high performance.

Tomato Classification 2024

Developed a tomato classification system on Raspberry Pi 5:

- Receives coordinate inputs and calculates the drawing paths by determining the start and end points of each edge.
- Raspberry Pi 5 runs a YOLO model to classify tomato types and control two servo motors for sorting.
- Based on detection results, GPIO signals trigger Servo1 for direction and Servo2 for releasing one tomato into the proper bin.

Container Damage Detection 2024

Using ESP32-CAM and Raspberry Pi 5 to detect defects in containers

- ESP32-CAM captures images of tomatoes and streams them to Raspberry Pi 5 via HTTP.
- The Raspberry Pi runs a YOLO-based model to detect defects and classify damaged areas in real-time.
- Detection results are logged and used to support automated inspection workflows.

Object Counting – Object Detection 2024

Develop a Python method using sklearn for precise object localization:

- Implemented feature extraction techniques including Canny Edge and HOG.
- Applied classification algorithms such as KNN, Logistic Regression, and SVM for detection.
- Utilized sliding window approach combined with non-maximum suppression to optimize localization accuracy.
- Evaluated the method on multiple datasets (car, pipe, human) and tuned parameters to select the best-performing model.